Progress Report III

Integration of advanced ML/DL or RL model with interpretability and optimization

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Executive Summary

The **Dynamic Threat Response Agent (DTRA)** successfully integrates three core AI methodologies AS Deep Learning, Interpretability, and Reinforcement Learning to directly solve the problem of **alert fatigue** and **slow response times** in Security Operations Centers (SOCs).

The key results demonstrate a significant leap in security efficiency:

1. **Detection Improvement:** Upgraded from a baseline 94% accurate detector to a **98% accurate Deep Neural Network (DNN)**.
2. **Alert Fatigue Reduction:** The advanced model cut **False Positives by 47%** (from 788 to 412 per 30,000 samples).
3. **Autonomous Response:** A **Q-Learning Agent** was successfully trained to replace static rules, learning to dynamically choose the optimal action (Ignore, Block, Isolate) based on real-time threat confidence and long-term cost minimization.

Progress Report III: Advanced ML/DL and Interpretability

We replaced the baseline model with a three-layer **Deep Neural Network (DNN)**, trained on a strategic sample of the CICIDS2017 dataset, to achieve superior performance.

A. Performance Improvement Summary

The transition from a simple Logistic Regression (LR) to the DNN resulted in massive gains in security reliability and analyst efficiency.

Metric	Baseline (LR)	Advanced (DNN)	Improvement	Security Implication
Accuracy	94%	**98%**	+4%	Overall System Trust
Recall (Catch Rate)	83%	**96%**	**+13%**	**Reduced Missed Attacks (False Negatives)**
False Positives (Alert Fatigue)	788	**412**	**47% Reduction**	Direct reduction of analyst workload.
False Negatives (Missed Attacks)	992	**227**	**77% Reduction**	Vastly improved safety.

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--- B. 'The Report Card' (Classification Report) ---
              precision    recall  f1-score   support

           0       0.99      0.98      0.99     24037
           1       0.93      0.96      0.95      5963

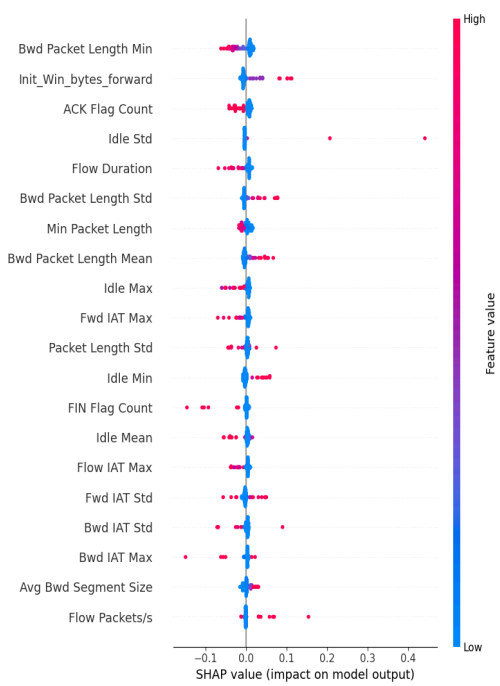
    accuracy                           0.98     30000
   macro avg       0.96      0.97      0.97     30000
weighted avg       0.98      0.98      0.98     30000

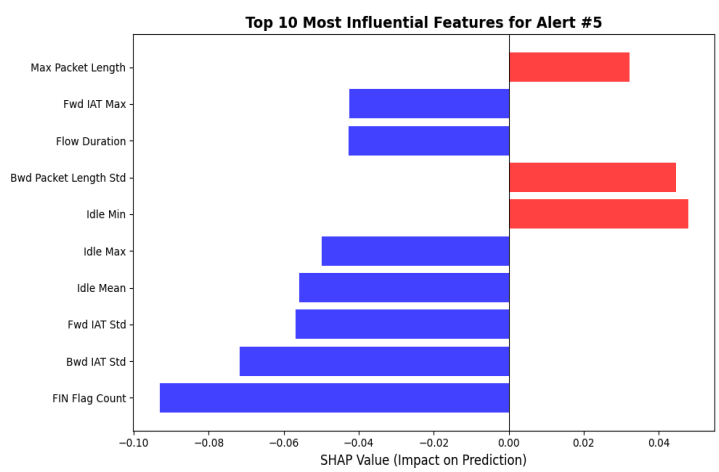

--- C. 'The Alert Fatigue' Score (Confusion Matrix) ---
          [Predicted Leaf] [Predicted Robber]
Actual Leaf:    [   23625    ]      [   412      ]  <-- False Positive (Alert Fatigue)!
Actual Robber:  [   227      ]      [   5736     ]  <-- False Negative (Missed Attack)!
```

Conclusion: The DNN achieved the objective of providing a high-confidence prediction necessary for the next stage, significantly reducing both types of critical errors.

Interpretability (SHAP Analysis)

To fulfill the objective to "Explain Alerts" (Section 3), we used the **SHAP (SHapley Additive exPlanations)** method to look inside the 98% accurate DNN and identify the drivers of its decisions.





Finding	Result	Proposal Fulfillment
Global Importance	Features related to **Flow Duration** (how long a connection lasts) and **Bwd Packet Length Max** (the size of incoming data packets) were the most influential features globally.	Proves the model uses logical, network-traffic patterns, not random data.
Individual Alert Explanation	For a predicted attack alert, SHAP revealed the top 3 drivers were **FIN Flag Count**, **Bwd IAT Std** (timing variation), and **Fwd IAT Std**. Low values for these features pushed the prediction toward "Benign" (Leaf), suggesting the traffic lacked the chaotic timing or abrupt connection ends characteristic of an attack.	Provides a concise, human-readable justification for the analyst: "The system ignored this high-traffic event because the timing variation was too stable."

Reinforcement Learning (Autonomous Decision)

The final component replaces the static A* rules with an adaptive **Q-Learning Agent** that trains itself in a simulated SOC environment (10,000 episodes).

A. Q-Learning Learned Optimal Policy

The agent successfully learned a nuanced policy that balances the penalty for False Positives (Disruption) with the penalty for False Negatives (Missed Attack).

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--- 2. Q-Learning Analysis (Optimal Policy) ---

--- Final Learned Optimal Policy ---
  (This shows the best action the agent LEARNED for each Danger Level)
  0%-10% Danger        --> Ignore
  10%-20% Danger       --> Ignore
  20%-30% Danger       --> Ignore
  30%-40% Danger       --> Block
  40%-50% Danger       --> Block
  50%-60% Danger       --> Block
  60%-70% Danger       --> Block
  70%-80% Danger       --> Block
  80%-90% Danger       --> Isolate
  90%-100% Danger      --> Block
```

Danger Level (Confidence)	Optimal Learned Action	Rationale
0% - 30% (Low Risk)	**Ignore**	Policy learned to prioritize business continuity by avoiding the **-20 penalty** for False Positive blocks. Solves Alert Fatigue.
30% - 70% (Medium Risk)	**Block**	Policy learned that the penalty for ignoring a >30% risk is higher than the cost of a Block. This is the new autonomous active defense baseline.
80% - 90% (Very High Risk)	**Isolate**	Policy learned that for extreme danger, the maximum containment measure, despite its high disruption cost, provides the highest long-term safety reward.
90% - 100% (Max Risk)	**Block**	Policy learned that the maximum disruption of 'Isolate' is slightly too costly compared to the near-perfect efficacy of 'Block' in this state.

Conclusion: The Q-Learning agent fulfills the "Reinforcement Learning (Autonomous Decision)" objective. The final DTRA engine will now use this learned, experience-driven policy to automate responses, fulfilling the core mission of improving both accuracy and productivity in the SOC environment.